

SHOOLINI UNIVERSITY

Practical Lab Assignment 01 — Git Version Control (Unit 1 & Unit 2)

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GitHub Repo:	https://github.com/datlaryan/github-assignment		

PROJECT 3: STUDENT CLUB MANAGEMENT REPOSITORY

Tasks 27 & 28: Initialize Repository and Configure User Details

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git init -b main
Initialized empty Git repository in Z:/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club/.git/

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git config user.name "Aryan"

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git config user.email "aryanpankaj@shooliniuniversity.com"

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git config --list --local
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.ignorecase=true
user.name=Aryan
user.email=aryanpankaj@shooliniuniversity.com

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ █
```

Tasks 29 & 30: Create Initial Club Files and Make First Commit

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git status
On branch main

No commits yet

Untracked files:
(use "git add <file>..." to include in what will be committed)
README.txt
club-info.txt
events.txt
members.txt

nothing added to commit but untracked files present (use "git add" to track)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git add .

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git commit -m "Initial student club repository"
[main (root-commit) 10ad11d] Initial student club repository
4 files changed, 15 insertions(+)
create mode 100644 README.txt
create mode 100644 club-info.txt
create mode 100644 events.txt
create mode 100644 members.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git log --oneline
10ad11d Initial student club repository

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ █
```

Task 31: Update README.txt and Inspect Staged/Unstaged Diffs

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git diff
diff --git a/README.txt b/README.txt
index e418823..495da3e 100644
--- a/README.txt
+++ b/README.txt
@@ -3,3 +3,9 @@
Welcome to the Student Club Management Repository.
This repository tracks club activities, member lists, and upcoming events.
+
+Club Purpose:
+To bridge the gap between academic curriculum and practical industry skills.
+
+Club Objectives:
+1. Conduct weekly coding and version control workshops.
+2. Organize annual hackathons and tech symposiums.
+3. Build open-source community projects.
+4. Mentor junior students in software engineering practices.

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git add README.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git diff --staged
diff --git a/README.txt b/README.txt
index e418823..495da3e 100644
--- a/README.txt
+++ b/README.txt
@@ -3,3 +3,9 @@
Welcome to the Student Club Management Repository.
This repository tracks club activities, member lists, and upcoming events.
+
+Club Purpose:
+To bridge the gap between academic curriculum and practical industry skills.
+
+Club Objectives:
+1. Conduct weekly coding and version control workshops.
+2. Organize annual hackathons and tech symposiums.
+3. Build open-source community projects.
+4. Mentor junior students in software engineering practices.

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git commit -m "Updated README with club purpose and objectives"
[main 9935b3d] Updated README with club purpose and objectives
1 file changed, 9 insertions(+)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$
```

Task 32: Create events Branch and Commit Proposed Events

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (events)
$ git switch -c events
Switched to a new branch 'events'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (events)
$ git add events.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (events)
$ git commit -m "Added proposed club events"
[events 930667f] Added proposed club events
1 file changed, 5 insertions(+)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (events)
$ git branch
* events
main

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (events)
$
```

Task 33: Create members Branch and Commit Club Members

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (events)
$ git switch main
Switched to branch 'main'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git switch -c members
Switched to a new branch 'members'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (members)
$ git add members.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (members)
$ git commit -m "Added sample club members"
[members 7657aa4] Added sample club members
1 file changed, 7 insertions(+)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (members)
$ git branch
events
main
* members

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (members)
$
```

Task 34: Create activities Branch and Commit Past Activities

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (members)
$ git switch main
Switched to branch 'main'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git switch -c activities
Switched to a new branch 'activities'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (activities)
$ git add club-info.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (activities)
$ git commit -m "Added previous club activities"
[activities dc67899] Added previous club activities
1 file changed, 5 insertions(+)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (activities)
$ git branch
* activities
events
main
members

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (activities)
$
```

Task 35: Merge events (Fast-Forward) and members (Three-Way) into main

```
MINGW64:/z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (activities)
$ git switch main
Switched to branch 'main'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git merge events
Updating 9935b3d..930667f
Fast-forward
 events.txt | 5 ++++
 1 file changed, 5 insertions(+)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git merge members -m "Merge branch 'members' into main"
Merge made by the 'ort' strategy.
 members.txt | 7 ++++++
 1 file changed, 7 insertions(+)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git log --oneline --graph --all
* 0a9bcf7 Merge branch 'members' into main
|\
| * 7657aa4 Added sample club members
* | 930667f Added proposed club events
|/
| * dc67899 Added previous club activities
|/
* 9935b3d Updated README with club purpose and objectives
* 10ad11d Initial student club repository

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$
```

Task 36: Create Intentional Merge Conflict in club-info.txt

```
MINGW64:/z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git switch activities
Switched to branch 'activities'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (activities)
$ git commit -am "Updated mission statement for AI focus on activities branch"
[activities 48f0caf] Updated mission statement for AI focus on activities branch
1 file changed, 1 insertion(+), 1 deletion(-)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (activities)
$ git switch main
Switched to branch 'main'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git commit -am "Updated mission statement for web and cloud on main branch"
[main 169ac9b] Updated mission statement for web and cloud on main branch
1 file changed, 3 insertions(+), 1 deletion(-)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git merge activities
Auto-merging club-info.txt
CONFLICT (content): Merge conflict in club-info.txt
Automatic merge failed; fix conflicts and then commit the result.

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main|MERGING)
$ git status
On branch main
You have unmerged paths.
(fix conflicts and run "git commit")
(use "git merge --abort" to abort the merge)

Unmerged paths:
(use "git add <file>..." to mark resolution)
both modified: club-info.txt

no changes added to commit (use "git add" and/or "git commit -a")

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main|MERGING)
$
```

Task 37: Resolve Conflict in club-info.txt and Finalize Merge Commit

```
MINGW64:/z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git diff
diff --cc club-info.txt
index 3739b42,20cecb5..0000000
--- a/club-info.txt
+++ b/club-info.txt
@@@ -2,6 -2,9 +2,15 @@@
Club Name: Shoolini Tech Innovators Club
Faculty Coordinator: Dr. Sharma
Student Lead: Aryan
++<<<<<< HEAD
+Mission: To empower students by mastering full-stack web engineering, cloud computing, and industry certifications.
++=====
+ Mission: To empower student developers through hands-on artificial intelligence projects and open source innovation.
++>>>>>> activities

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git add club-info.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git commit -m "Resolved merge conflict in club-info mission statement"
[main 8497325] Resolved merge conflict in club-info mission statement

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git status
On branch main
nothing to commit, working tree clean

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ █
```

Task 38: Create Annotated Tag v1.0 and View Tag Details

```
MINGW64:/z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git tag -a v1.0 -m "Student Club Management v1.0 Release"

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git tag
v1.0

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git show v1.0
tag v1.0
Tagger: Aryan <aryanpankaj@shooliniuniversity.com>
Date: Fri Sep 25 20:16:27 2026 +0530

Student Club Management v1.0 Release

commit 8497325fe833604914d681fcd7c771b45ed2925b
Merge: 169ac9b 48f0caf
Author: Aryan <aryanpankaj@shooliniuniversity.com>
Date: Fri Sep 25 20:16:26 2026 +0530

Resolved merge conflict in club-info mission statement

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ █
```

Task 39: Demonstrate Undo Operation using git restore on events.txt

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   events.txt

no changes added to commit (use "git add" and/or "git commit -a")

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git restore events.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git status
On branch main
nothing to commit, working tree clean

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ █
```

Task 40: Repository Audit and Final Commit Graph (git log --graph)

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git status
On branch main
nothing to commit, working tree clean

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git branch -a
activities
events
* main
members

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ git log --oneline --graph --all
* 8497325 Resolved merge conflict in club-info mission statement
|\
| * 48f0caf Updated mission statement for AI focus on activities branch
| * dc67899 Added previous club activities
* | 169ac9b Updated mission statement for web and cloud on main branch
* | 0a9bcf7 Merge branch 'members' into main
|\ \
| * | 7657aa4 Added sample club members
| | /
* / 930667f Added proposed club events
| /
* 9935b3d Updated README with club purpose and objectives
* 10ad11d Initial student club repository

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/student-club (main)
$ █
```

PROJECT 4: PERSONAL EXPENSE TRACKER

Tasks 42 & 43: Create expense-tracker Directory, Initialize Git, and Configure User

```
MINGW64:/z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git init -b main
Initialized empty Git repository in Z:/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker/.git/

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git config user.name "Aryan"

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git config user.email "aryanpankaj@shooliniuniversity.com"

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git config --list --local
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.ignorecase=true
user.name=Aryan
user.email=aryanpankaj@shooliniuniversity.com

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ █
```

Tasks 44 & 45: Create Initial Files and Baseline Commit

```
MINGW64:/z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git status
On branch main

No commits yet

Untracked files:
(use "git add <file>..." to include in what will be committed)
README.txt
categories.txt
expense.txt
summary.txt

nothing added to commit but untracked files present (use "git add" to track)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git add .

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git commit -m "Initial expense tracker"
[main (root-commit) 92459d7] Initial expense tracker
4 files changed, 17 insertions(+)
create mode 100644 README.txt
create mode 100644 categories.txt
create mode 100644 expense.txt
create mode 100644 summary.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git log --oneline
92459d7 Initial expense tracker

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ █
```

Task 46: Create food-expenses Branch and Record Food Expenses

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (food-expenses)
$ git switch -c food-expenses
Switched to a new branch 'food-expenses'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (food-expenses)
$ git add expense.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (food-expenses)
$ git commit -m "Added five food expenses"
[food-expenses af390ec] Added five food expenses
1 file changed, 5 insertions(+)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (food-expenses)
$ git branch
* food-expenses
main

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (food-expenses)
$
```

Task 47: Create travel-expenses Branch and Record Travel Expenses

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (food-expenses)
$ git switch main
Switched to branch 'main'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git switch -c travel-expenses
Switched to a new branch 'travel-expenses'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (travel-expenses)
$ git add expense.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (travel-expenses)
$ git commit -m "Added five travel expenses"
[travel-expenses eded946] Added five travel expenses
1 file changed, 5 insertions(+)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (travel-expenses)
$ git branch
food-expenses
main
* travel-expenses

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (travel-expenses)
$
```

Task 48: Create monthly-summary Branch and Add Summary Record

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (travel-expenses)
$ git switch main
Switched to branch 'main'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git switch -c monthly-summary
Switched to a new branch 'monthly-summary'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (monthly-summary)
$ git add summary.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (monthly-summary)
$ git commit -m "Added monthly expense summary"
[monthly-summary 39e2b32] Added monthly expense summary
1 file changed, 3 insertions(+), 2 deletions(-)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (monthly-summary)
$ git branch
food-expenses
main
* monthly-summary
travel-expenses

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (monthly-summary)
$
```

Task 49: View Divergent Branches and Commit History Before Merge

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (monthly-summary)
$ git switch main
Switched to branch 'main'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git branch -a
food-expenses
* main
monthly-summary
travel-expenses

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git log --oneline --graph --all
* 39e2b32 Added monthly expense summary
| * eded946 Added five travel expenses
|/
| * af390ec Added five food expenses
|/
* 92459d7 Initial expense tracker

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$
```

Task 50: Merge food-expenses and travel-expenses into main

```
MINGW64:/z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git merge food-expenses
Updating 92459d7..af390ec
Fast-forward
 expense.txt | 5 ++++
 1 file changed, 5 insertions(+)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git merge travel-expenses
Auto-merging expense.txt
CONFLICT (content): Merge conflict in expense.txt
Automatic merge failed; fix conflicts and then commit the result.

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git add expense.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git commit -m "Merged food and travel expenses into main"
[main bc55b84] Merged food and travel expenses into main

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git log --oneline --graph --all
* bc55b84 Merged food and travel expenses into main
|\
| * eded946 Added five travel expenses
* | af390ec Added five food expenses
|/
| * 39e2b32 Added monthly expense summary
|/
* 92459d7 Initial expense tracker

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ █
```

Task 51: Create Intentional Conflict in summary.txt on monthly-summary Merge

```
MINGW64:/z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git switch monthly-summary
Switched to branch 'monthly-summary'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (monthly-summary)
$ git commit -am "Updated total expenses calculation on monthly-summary"
[monthly-summary 09ed111] Updated total expenses calculation on monthly-summary
1 file changed, 2 insertions(+), 3 deletions(-)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (monthly-summary)
$ git switch main
Switched to branch 'main'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git commit -am "Updated audited total expenses on main branch"
[main 2b1fb11] Updated audited total expenses on main branch
1 file changed, 2 insertions(+), 2 deletions(-)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git merge monthly-summary
Auto-merging summary.txt
CONFLICT (content): Merge conflict in summary.txt
Automatic merge failed; fix conflicts and then commit the result.

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main|MERCING)
$ git status
On branch main
You have unmerged paths.
(fix conflicts and run "git commit")
(use "git merge --abort" to abort the merge)

Unmerged paths:
(use "git add <file>..." to mark resolution)
both modified: summary.txt

no changes added to commit (use "git add" and/or "git commit -a")

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main|MERCING)
$ █
```

Task 52: Manually Resolve summary.txt Conflict and Commit

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git diff
diff --cc summary.txt
index a6e9a27,f14d747..0000000
--- a/summary.txt
+++ b/summary.txt
@@@ -1,4 -1,4 +1,9 @@@
=== Monthly Expense Summary: September 2026 ===
Total Monthly Budget: Rs. 15,000
++<<<<<< HEAD
+Total Recorded Expenses: Rs. 5,100 (Audited real expenses verified on main branch)
++=====
+ Total Recorded Expenses: Rs. 4,800 (Calculated by Budget Analyst on monthly-summary)
++>>>>>> monthly-summary
aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git add summary.txt
aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git commit -m "Resolved merge conflict in monthly summary"
[main cb457d9] Resolved merge conflict in monthly summary
aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git status
On branch main
nothing to commit, working tree clean
aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$
```

Task 53: Create Annotated Tag v1.0 and Verify Metadata

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git tag -a v1.0 -m "Personal Expense Tracker v1.0 Release"
aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git tag
v1.0
aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git show v1.0
tag v1.0
Tagger: Aryan <aryanpankaj@shooliniuniversity.com>
Date: Fri Sep 25 20:18:00 2026 +0530

Personal Expense Tracker v1.0 Release

commit cb457d919e75a7be3d9cf0e2128d4b20b05c81b9
Merge: 2b1fb11 09ed111
Author: Aryan <aryanpankaj@shooliniuniversity.com>
Date: Fri Sep 25 20:17:59 2026 +0530

Resolved merge conflict in monthly summary
aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$
```

Task 54: Make Incorrect Change in categories.txt and Discard with git restore

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   categories.txt

no changes added to commit (use "git add" and/or "git commit -a")

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git restore categories.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git status
On branch main
nothing to commit, working tree clean

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$
```

Task 55: Demonstrate git diff and git diff --staged Before Commit

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git diff
diff --git a/categories.txt b/categories.txt
index bcc219f..15442b2 100644
--- a/categories.txt
+++ b/categories.txt
@@ -4,3 +4,5 @@
 3. Books & Education
 4. Utilities & Bills
 5. Entertainment & Leisure
+6. Healthcare & Medical
+7. Investments & Savings

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git add categories.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git diff --staged
diff --git a/categories.txt b/categories.txt
index bcc219f..15442b2 100644
--- a/categories.txt
+++ b/categories.txt
@@ -4,3 +4,5 @@
 3. Books & Education
 4. Utilities & Bills
 5. Entertainment & Leisure
+6. Healthcare & Medical
+7. Investments & Savings

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git commit -m "Added Healthcare and Investment expense categories"
[main 4ac3b09] Added Healthcare and Investment expense categories
1 file changed, 2 insertions(+)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$
```

Task 56: Final Commit Graph and Clean Working Tree Status

```
MINGW64:/z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git status
On branch main
nothing to commit, working tree clean

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git branch -a
food-expenses
* main
monthly-summary
travel-expenses

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$ git log --oneline --graph --all
* 4ac3b09 Added Healthcare and Investment expense categories
* cb457d9 Resolved merge conflict in monthly summary
|\
| * 09ed111 Updated total expenses calculation on monthly-summary
| * 39e2b32 Added monthly expense summary
* | 2b1fb11 Updated audited total expenses on main branch
* | bc55b84 Merged food and travel expenses into main
|\ \
| * | eded946 Added five travel expenses
| | /
* / af390ec Added five food expenses
| /
* 92459d7 Initial expense tracker

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/expense-tracker (main)
$
```

PROJECT 5: MINI LIBRARY MANAGEMENT SYSTEM

Tasks 58 & 59: Initialize library-management Repository and Verify User Config

```
MINGW64:/z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git init -b main
Initialized empty Git repository in Z:/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management/.git/

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git config user.name "Aryan"

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git config user.email "aryanpankaj@shooliniuniversity.com"

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git config --list --local
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.ignorecase=true
user.name=Aryan
user.email=aryanpankaj@shooliniuniversity.com

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ █
```

Tasks 60 & 61: Create Catalog Files and Commit Initial Version

```
MINGW64:/z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git status
On branch main

No commits yet

Untracked files:
(use "git add <file>..." to include in what will be committed)
README.txt
books.txt
issued-books.txt
members.txt

nothing added to commit but untracked files present (use "git add" to track)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git add .

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git commit -m "Initial library management project"
[main (root-commit) 9004d4c] Initial library management project
4 files changed, 14 insertions(+)
create mode 100644 README.txt
create mode 100644 books.txt
create mode 100644 issued-books.txt
create mode 100644 members.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git log --oneline
9004d4c Initial library management project

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ █
```

Task 62: Update books.txt with Editions/Publishers as Second Version

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git diff
diff --git a/books.txt b/books.txt
index 42359c5..67ba940 100644
--- a/books.txt
+++ b/books.txt
@@ -1,5 +1,5 @@
=== Library Catalog ===
-[Book ID] [Title] [Author] [Status]
-B101 Introduction to Algorithms Cormen et al. Available
-B102 Clean Code Robert C. Martin Available
-B103 Operating System Concepts Silberschatz Available
+[Book ID] [Title] [Author] [Edition / Publisher] [Status]
+B101 Introduction to Algorithms Cormen et al. 4th Ed (MIT Press) Available
+B102 Clean Code Robert C. Martin 1st Ed (Prentice Hall) Available
+B103 Operating System Concepts Silberschatz 10th Ed (Wiley) Available

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git add books.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git commit -m "Updated books catalog with editions and publishers"
[main 1ad17d5] Updated books catalog with editions and publishers
1 file changed, 4 insertions(+), 4 deletions(-)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ █
```

Task 63: Create books-update Branch and Add Computer Science Books

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (books-update)
$ git switch -c books-update
Switched to a new branch 'books-update'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (books-update)
$ git add books.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (books-update)
$ git commit -m "Added five additional computer science books to catalog"
[books-update ea34d36] Added five additional computer science books to catalog
1 file changed, 5 insertions(+)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (books-update)
$ git branch
* books-update
main

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (books-update)
$ █
```

Task 64: Create members-update Branch and Add Library Members

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (books-update)
$ git switch main
Switched to branch 'main'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git switch -c members-update
Switched to a new branch 'members-update'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (members-update)
$ git add members.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (members-update)
$ git commit -m "Added five new registered library members"
[members-update 0075d2f] Added five new registered library members
1 file changed, 5 insertions(+)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (members-update)
$ git branch
books-update
main
* members-update

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (members-update)
$
```

Task 65: Create issue-records Branch and Add Borrowing Transactions

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (members-update)
$ git switch main
Switched to branch 'main'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git switch -c issue-records
Switched to a new branch 'issue-records'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (issue-records)
$ git add issued-books.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (issue-records)
$ git commit -m "Added five book issue and return records"
[issue-records da4be33] Added five book issue and return records
1 file changed, 5 insertions(+)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (issue-records)
$ git branch
books-update
* issue-records
main
members-update

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (issue-records)
$
```

Task 66: View Branches and Commit History Before Merging

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (issue-records)
$ git switch main
Switched to branch 'main'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git branch -a
books-update
issue-records
* main
members-update

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git log --oneline --graph --all
* da4be33 Added five book issue and return records
| * 0075d2f Added five new registered library members
|/
| * ea34d36 Added five additional computer science books to catalog
|/
* 1ad17d5 Updated books catalog with editions and publishers
* 9004d4c Initial library management project

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$
```

Task 67: Fast-Forward Merge (books-update) vs. Three-Way Merge (members-update)

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git merge books-update
Updating 1ad17d5..ea34d36
Fast-forward
 books.txt | 5 +++++
1 file changed, 5 insertions(+)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git merge members-update -m "Merge branch 'members-update' into main"
Merge made by the 'ort' strategy.
 members.txt | 5 +++++
1 file changed, 5 insertions(+)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git log --oneline --graph --all
* 09002d1 Merge branch 'members-update' into main
| \
| * 0075d2f Added five new registered library members
* | ea34d36 Added five additional computer science books to catalog
|/
| * da4be33 Added five book issue and return records
|/
* 1ad17d5 Updated books catalog with editions and publishers
* 9004d4c Initial library management project

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$
```

Task 68: Create Intentional Conflict on Record ISS-1001 in issued-books.txt

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git switch issue-records
Switched to branch 'issue-records'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (issue-records)
$ git commit -am "Updated ISS-1001 to Returned on issue-records branch"
[issue-records 12b0fe8] Updated ISS-1001 to Returned on issue-records branch
1 file changed, 1 insertion(+), 1 deletion(-)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (issue-records)
$ git switch main
Switched to branch 'main'

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git commit -am "Updated ISS-1001 due date extension on main branch"
[main f79abef] Updated ISS-1001 due date extension on main branch
1 file changed, 1 insertion(+), 1 deletion(-)

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git merge issue-records
Auto-merging issued-books.txt
CONFLICT (content): Merge conflict in issued-books.txt
Automatic merge failed; fix conflicts and then commit the result.

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main|MERGING)
$ git status
On branch main
You have unmerged paths.
(fix conflicts and run "git commit")
(use "git merge --abort" to abort the merge)

Unmerged paths:
(use "git add <file>..." to mark resolution)
both modified: issued-books.txt

no changes added to commit (use "git add" and/or "git commit -a")

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main|MERGING)
$ █
```

Task 69: Resolve Conflict on Record ISS-1001 and Complete Merge

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git diff
diff --cc issued-books.txt
index 5ddd953,a5702ca..0000000
--- a/issued-books.txt
+++ b/issued-books.txt
@@@ -1,3 -1,8 +1,12 @@@
=== Book Issue and Return Log ===
[Issue ID] [Book ID] [Member ID] [Issue Date] [Due Date] [Return Status]
++<<<<<< HEAD
+ISS-1001 B101 M001 2026-09-01 2026-09-30 Due Date Extended on Main (Special Permission)
++=====
+ ISS-1001 B101 M001 2026-09-01 2026-09-15 Returned (Condition: Excellent, Fine: Rs. 0)
++>>>>>> issue-records

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git add issued-books.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git commit -m "Resolved merge conflict in issued-books record ISS-1001"
[main 2a19f2c] Resolved merge conflict in issued-books record ISS-1001

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git status
On branch main
nothing to commit, working tree clean

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ █
```

Task 70: Create Annotated Release Tag v1.0 and Verify Metadata

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git tag -a v1.0 -m "Mini Library Management System v1.0 Release"

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git tag
v1.0

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git show v1.0
tag v1.0
Tagger: Aryan <aryanpankaj@shooliniuniversity.com>
Date: Fri Sep 25 20:18:34 2026 +0530

Mini Library Management System v1.0 Release

commit 2a19f2c5c7d20a3ec0300434edb4c1a6245f7011
Merge: f79abef 12b0fe8
Author: Aryan <aryanpankaj@shooliniuniversity.com>
Date: Fri Sep 25 20:18:32 2026 +0530

Resolved merge conflict in issued-books record ISS-1001

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ █
```

Task 71: Discard Unwanted Corruption in books.txt using git restore

```
MINGW64/z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    modified:   books.txt

no changes added to commit (use "git add" and/or "git commit -a")

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git restore books.txt

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git status
On branch main
nothing to commit, working tree clean

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ █
```

Task 72: Final Verification: Status, Diff, Log Graph, and Branch List

```
MINGW64:/z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git status
On branch main
nothing to commit, working tree clean

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git branch -a
books-update
issue-records
* main
members-update

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$ git log --oneline --graph --all
* 2a19f2c Resolved merge conflict in issued-books record ISS-1001
|\
| * 12b0fe8 Updated ISS-1001 to Returned on issue-records branch
| * da4be33 Added five book issue and return records
* | f79abef Updated ISS-1001 due date extension on main branch
* | 09002d1 Merge branch 'members-update' into main
|\ \
| * | 0075d2f Added five new registered library members
| | /
* / ea34d36 Added five additional computer science books to catalog
| /
* 1ad17d5 Updated books catalog with editions and publishers
* 9004d4c Initial library management project

aryan@MORTIS2K MINGW64 /z/Shoolini/github-sem-3/ASSIGNMENT - 01/library-management (main)
$
```

QUESTIONS AND ANSWERS (ASSIGNMENT REVIEW)

Task 74: Write a short explanation of Fast-Forward versus Three-Way merge based on your project.

1. Fast-Forward Merge: Occurs when the target branch (**main**) has not received any new commits since the feature branch diverged. Git simply advances the target branch pointer directly to the tip commit of the incoming branch. No merge commit is created, and commit history remains strictly linear.

Demonstrated in Project 5 (Task 67): Merging **books-update** into **main** executed as a fast-forward merge because **main** had no intermediate commits.

2. Three-Way Merge: Occurs when the target branch and incoming branch have diverged—both have accumulated independent commits after their common ancestor. Git evaluates three commits (Common Ancestor, Current HEAD, and Incoming Tip) to synthesize changes. Disjoint modifications are merged automatically, and a dedicated **Merge Commit** with two parent commits is generated.

Demonstrated in Project 5 (Task 67): Merging **members-update** after **main** had already incorporated **books-update** forced a three-way merge because **main** had moved ahead.

Property	Fast-Forward Merge	Three-Way Merge
Topology	Linear (Target is direct ancestor of feature branch)	Divergent (Independent commits on both branches)
Merge Commit	No merge commit created	Creates an explicit merge commit with 2 parents
Conflict Risk	None (no conflicting base states)	Possible if overlapping lines were modified
History Shape	Single straight line	Branched tree with visual merge join nodes

Question: What are Git conflict markers and how do you resolve a merge conflict?

When Git cannot automatically reconcile changes to the same line in two branches, it halts and marks the conflicting file with conflict markers:

- <<<<<<< **HEAD**: Indicates changes present on the current active branch.
- =====: Separates the conflicting content from both branches.
- >>>>>>> [**branch-name**]: Indicates incoming changes from the branch being merged.

Resolution Process:

1. Open the conflicted file in an editor and manually edit the lines to retain the correct combined information.
2. Delete all marker lines (<<<<<<<, =====, >>>>>>>).
3. Stage the resolved file using `git add <file>`.
4. Finalize the merge by executing `git commit`.

Question: What is the operational difference between 'git diff' and 'git diff --staged'?

git diff: Compares modifications in the working directory against the staging area (Git index). It reveals changes that have been made to tracked files but not yet staged with `git add`.

git diff --staged (or `git diff --cached`): Compares staged modifications in the Git index against the most recent commit (**HEAD**). It displays the exact patch that will be included in the next commit.

Question: How does 'git restore' safely discard changes?

`git restore <file>` copies the clean version of a file from the Git index (or **HEAD**) into the working tree, discarding any unstaged edits without affecting commit history or staged changes. This provides a safe, targeted undo mechanism for uncommitted working tree modifications.